

Version 1 - Baseline XGBoost - Feature Used / Dropped Audit

เทียบ source clean dataset กับ df_engineered และระบุว่า feature ใดใช้ train model / feature ใดตัดทิ้ง

Model Summary

Model	Dataset	df_engineered	Used Features	Accuracy	Recall	F1	AUC	Rating
XGBoost baseline	clean_dataset.csv	df_engineered.csv	136	70.80%	26.80%	34.82%	68.82%	D

Source Artifacts

artifact	path	columns	sample_rows_read
source_data	data\processed\clean_dataset.csv	65	5
df_engineered	docs\version 1\data\features\df_engineered.csv	51	5
used_feature_list	docs\version 1\data\features\train_test_sets.pkl	136	pkl

Used Feature Groups

group	feature_count	example_features
product	40	category_Electronics, category_Fashion, category_Home_Appliance, category_Supplement, category_channel_Cosmetics_Shopee
geography	29	gender_province_Female_Chiang_Mai, gender_province_Female_Chonburi, gender_province_Female_Khon_Kaen, gender_province_Female_Nonthaburi, gender_province_Female_Phuket
order_channel_payment	28	channel_type_Shopee, channel_type_TV_Show, channel_type_TikTok, payment_method_COD, payment_method_Credit_Card
other	13	customer_return_ratio, is_cod, is_fashion_tv, is_first_order, is_high_risk_customer
customer_history	10	hist_order_count_180d, hist_order_count_30d, hist_order_count_60d, hist_return_rate_180d, hist_return_rate_30d
time	8	customer_age_days, days_since_last_order, days_since_last_return, delivery_time_expected_days, is_peak_hour
price_discount	6	is_high_discount, log_total_amount, log_unit_price, promo_discount_pct, total_amount
customer_profile	2	age, membership_tier

Used Features (sample)

age, category_Electronics, category_Fashion, category_Home_Appliance, category_Supplement, category_channel_Cosmetics_Shopee, category_channel_Cosmetics_TV_Show, category_channel_Cosmetics_TikTok, category_channel_Electronics_Mobile_App, category_channel_Electronics_Shopee, category_channel_Electronics_TV_Show, category_channel_Electronics_TikTok, category_channel_Fashion_Mobile_App, category_channel_Fashion_Shopee, category_channel_Fashion_TV_Show, category_channel_Fashion_TikTok, category_channel_Home_Appliance_Mobile_App, category_channel_Home_Appliance_Shopee, category_channel_Home_Appliance_TV_Show, category_channel_Home_Appliance_TikTok, category_channel_Supplement_Mobile_App, category_channel_Supplement_Shopee, category_channel_Supplement_TV_Show, category_channel_Supplement_TikTok, category_payment_Cosmetics_COD, category_payment_Cosmetics_Credit_Card, category_payment_Electronics_Bank_Transfer, category_payment_Electronics_COD, category_payment_Electronics_Credit_Card, category_payment_Fashion_Bank_Transfer, category_payment_Fashion_COD, category_payment_Fashion_Credit_Card, category_payment_Home_Appliance_Bank_Transfer, category_payment_Home_Appliance_COD, category_payment_Home_Appliance_Credit_Card, category_payment_Supplement_Bank_Transfer, category_payment_Supplement_COD, category_payment_Supplement_Credit_Card, channel_type_Shopee, channel_type_TV_Show, channel_type_TikTok, customer_age_days, customer_return_ratio, days_since_last_order, days_since_last_return, delivery_time_expected_days, gender_province_Female_Chiang_Mai, gender_province_Female_Chonburi, gender_province_Female_Khon_Kaen, gender_province_Female_Nonthaburi, gender_province_Female_Phuket, gender_province_Female_Remote_Area, gender_province_Female_Songkhla, gender_province_Male_Bangkok, gender_province_Male_Chiang_Mai, gender_province_Male_Chonburi, gender_province_Male_Khon_Kaen, gender_province_Male_Nonthaburi, gender_province_Male_Phuket, gender_province_Male_Remote_Area ...

Dropped / Not Used Feature Logic

- คัด leakage/post-event/target fields จำนวนประมาณ 17 feature เช่น avg_delivery_days, delay_days, delivery_date, delivery_days, expected_delivery_date, is_returned, item_condition, refund_amount, return_date, return_id
- คัด identifier/high-cardinality fields จำนวนประมาณ 13 feature เช่น courier_id, courier_name, customer_id, customer_name, customer_phone, order_id, product_id, product_name, promo_id, promo_name
- feature ที่เป็น raw date หรือ raw text จะถูกแปลงเป็น numeric/encoded/rolling feature ก่อน หรือไม่ใช่ถ้าไม่ช่วย performance
- feature ที่เพิ่มแล้ว performance ไม่ดีขึ้นหรือซ้ำซ้อนกับ feature สำคัญ จะถูกจัดเป็น NOT USED เพื่อลด resource/query cost

Feature Audit Detail (sample rows)

Feature	In Clean	In Engineered	Used	Status	Reason
age	True	True	True	USED	model input feature
category_Electronics	False	False	True	USED	model input feature
category_Fashion	False	False	True	USED	model input feature
category_Home_Appliance	False	False	True	USED	model input feature
category_Supplement	False	False	True	USED	model input feature
category_channel_Cosmetics_Shopee	False	False	True	USED	model input feature
category_channel_Cosmetics_TV_Show	False	False	True	USED	model input feature
category_channel_Cosmetics_TikTok	False	False	True	USED	model input feature
category_channel_Electronics_Mobile_App	False	False	True	USED	model input feature
category_channel_Electronics_Shopee	False	False	True	USED	model input feature
category_channel_Electronics_TV_Show	False	False	True	USED	model input feature
category_channel_Electronics_TikTok	False	False	True	USED	model input feature
avg_delivery_days	True	False	False	DROPPED	post-event/target/leakage field
brand	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
campaign_discount_pct	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
category	True	True	False	NOT USED	not selected in final feature set
category_channel	False	True	False	NOT USED	engineered candidate not selected for this model
category_payment	False	True	False	NOT USED	engineered candidate not selected for this model
channel_type	True	True	False	NOT USED	not selected in final feature set
courier_id	True	False	False	DROPPED	identifier or high-cardinality identity field
courier_name	True	False	False	DROPPED	identifier or high-cardinality identity field
courier_type	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
coverage_region	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
customer_id	True	False	False	DROPPED	identifier or high-cardinality identity field
customer_name	True	False	False	DROPPED	identifier or high-cardinality identity field
customer_phone	True	False	False	DROPPED	identifier or high-cardinality identity field
damage_rate	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
delay_days	True	False	False	DROPPED	post-event/target/leakage field
delivery_date	True	False	False	DROPPED	post-event/target/leakage field
delivery_days	True	False	False	DROPPED	post-event/target/leakage field

